

# YIHAN WANG (王艺涵)

Department of Astronomy  
University of Wisconsin, Madison  
500 Lincoln Dr, Madison, WI 53706

Email: [wang3697@wisc.edu](mailto:wang3697@wisc.edu)  
[Personal Homepage](#)  
[Google Scholar Profile](#)

## EDUCATION

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|------------------------------------------------------------------------------------------------------|-------------|
| <b>Ph.D. in Physics</b><br>Stony Brook University, NY<br><i>Advisor: Rosalba Perna</i>               | 2017 - 2022 |
| <b>M.A. in Physics</b><br>Stony Brook University, NY                                                 | 2016 - 2017 |
| <b>B.S. in Physics (Theoretical Physics)</b><br>University of Science and Technology of China (USTC) | 2011 - 2015 |

## PROFESSIONAL APPOINTMENTS

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|----------------------------------------------------------------------------------|----------------|
| <b>Postdoctoral Research Associate</b><br>University of Wisconsin, Madison       | 2025 - Present |
| <b>Fellow, Nevada Center for Astrophysics</b><br>University of Nevada, Las Vegas | 2022 - 2025    |

## RESEARCH INTERESTS

### Cluster Dynamics and Multi-messenger Astrophysics

My research develops a unified theoretical framework for the coupled evolution of accretion disks and stellar populations in galactic nuclei. I investigate how dissipative disk processes actively regulate few-body interactions, population evolution, and transient production. By bridging small-scale gravitational dynamics with large-scale disk-mediated evolution, I study how dense nuclear environments generate electromagnetic transients and gravitational wave sources.

- **AGN Disk-Cluster Coupling:** Stellar and compact object dynamics regulated by AGN disks, including capture, migration, gas-assisted scattering, and binary formation.
- **Nuclear Transients:** Disk-assisted tidal disruption events, changing-look AGN, and quasi-periodic eruptions driven by disk-star interactions.
- **Gravitational Wave Sources:** Formation and evolution of extreme mass-ratio inspirals and binary black holes in disk-embedded nuclear star clusters.

## GRANTS & COMPUTING ALLOCATIONS

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| <b>ACCESS-SDSC   1.25M CPU hours</b><br>Project: <i>Propagation of Fast Radio Bursts</i> (PI)            | 2023 - 2025 |
| <b>ACCESS-SDSC   0.5M CPU hours</b><br>Project: <i>Hot Jupiter Formation in Dense Star Clusters</i> (PI) | 2022 - 2024 |

## HONORS & AWARDS

|                                                      |            |
|------------------------------------------------------|------------|
| Gerald Brown Prize                                   | 2022       |
| Di Tian Prize for Excellent Research                 | 2020       |
| National Endeavor Scholarship                        | 2014       |
| Outstanding Student Scholarship (USTC)               | 2012, 2013 |
| Exceptional Prize, Physics Experiment Contest (USTC) | 2013       |

## PUBLICATIONS

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A complete and updated publication list is available on [NASA ADS](#) and [Google Scholar](#).

- [30] **Wang, Y.**, Chen C., Zhang B., *VegasAfterglow: A High-Performance Framework for Gamma-Ray Burst Afterglows*, 2025, submitted to JHEAP, [arXiv.2507.10829](#)
- [29] Chen C., **Wang Y.**, Zhang B., *X-ray Emission Signatures of Neutron Star Mergers*, 2025, submitted to ApJ, [arXiv.2505.01606](#)
- [28] Li W.-X., ..., **Wang Y.-H.**, ..., *An extremely soft and weak fast X-ray transient associated with a luminous supernova*, 2025, submitted, [arXiv.2504.17034](#)
- [27] **Wang Y.**, Perna R., Zhu Z., Lin D. N. C., *Evolution of Extremely Soft Binaries in Dense Star Clusters: On the Jupiter Mass Binary Objects*, 2024, submitted to ApJ, [arXiv.2310.00020](#)
- [26] **Wang Y.**, Graham M., Ford S., McKernan B., Ryu T., Stern D., *Conditions for Changing-Look AGNs from Accretion Disk-Induced Tidal Disruption Events*, 2024, submitted to ApJL, [arXiv:2406.12096](#)
- [25] **Wang Y.**, Zhang B., *Evidence of a Past Merger of the Galactic Center Black Hole*, 2024, [Nature Astronomy](#)
- [24] **Wang Y.**, Perna R., Zhu Z., *Floating binary planets from ejections during close stellar encounters*, 2024, [Nature Astronomy](#)
- [23] **Wang Y.**, Lin D. N. C., Zhang B., Zhu Z., *Changing-Look AGN Behaviour Induced by Disk-Captured Tidal Disruption Events*, 2023, [ApJL](#), **962**, L7
- [22] Prasad C., **Wang Y.**, Perna R., Ford S., McKernan B., *Tidal Disruption Events from three-body scatterings in the disks of Active Galactic Nuclei*, 2023, submitted to MNRAS, [arXiv.2310.00020](#)
- [21] **Wang Y.**, Zhang B., Zhu Z., *Anisotropic Energy Injection from Magnetar Central Engines in Short GRBs*, 2023, [MNRAS](#), **528**, 3705
- [20] **Wang Y.**, Zhu Z., Lin D. N. C., *Stellar/BH Population in AGN Disks: Direct Binary Formation from Capture Objects in Nuclei Clusters*, 2023, [MNRAS](#), **528**, 4958
- [19] **Wang Y.**, Ford S., Perna R., McKernan B., Zhu Z., Zhang B., *Effective two-body scatterings around a massive object*, 2023, [MNRAS](#), **523**, 2014
- [18] Xin C., Haiman Z., Perna R., **Wang Y.**, Ryu T., *Tidal Peeling Events: low-eccentricity tidal disruption of a star by a stellar-mass black hole*, 2023, [ApJ](#), **961**, 149
- [17] **Wang Y.-H.**, Lazzati D., Perna R., *The emergence of diffused Gamma-Ray Burst afterglows from the disks of Active Galactic Nuclei*, 2022, [MNRAS](#), **516**, 5935
- [16] Ryu, T., Perna R., **Wang Y.-H.**, *Close Encounters of Stars with Stellar-mass Black Hole Binaries*, 2022, [MNRAS](#), **516**, 2204
- [15] Perna R., Artale M. C., **Wang Y.-H.**, et al., *Host galaxies and electromagnetic counterparts to binary neutron star mergers across the cosmic time*, 2021, [MNRAS](#), **512**, 2654
- [14] **Wang Y.-H.**, Perna R., Leigh N. W. C., Shara M. M., *Hot Jupiter formation in star cluster: Secular chaos*, 2021, [MNRAS](#), **509**, 5053
- [13] **Wang Y.-H.**, McKernan B., Ford S., Perna R., Leigh N. W. C., Mac Low M.-M., *Symmetry Breaking in Dynamical Encounters in the Disks of Active Galactic Nuclei*, 2021, [ApJL](#), **923**, L23
- [12] **Wang Y.-H.**, Leigh N. W. C., Liu B., Perna R., *SpaceHub: A high-performance gravity integration toolkit for few-body problems in astrophysics*, 2021, [MNRAS](#), **505**, 1053
- [11] **Wang Y.-H.**, Perna R., Armitage P. J., *Partial tidal disruption events by stellar mass black holes: Gravitational instability of stream and impact from remnant core*, 2021, [MNRAS](#), **503**, 6005

- [10] **Wang Y.-H.**, Leigh N. W. C., Perna R., Shara M. M., *Hot Jupiter and ultra-cold Saturn formation in dense star clusters*, 2020, [ApJ](#), **905**, 136
- [9] **Wang Y.-H.**, Perna R., Leigh N. W. C., *Planetary architectures in interacting stellar environments*, 2020, [MNRAS](#), **496**, 1453
- [8] **Wang Y.-H.**, Perna R., Leigh N. W. C., *Giant Planet Swaps during Close Stellar Encounters*, 2020, [ApJL](#), **891**, L14
- [7] **Wang Y.-H.**, Leigh N. W. C., Sesana A., Perna R., *The cosmological distribution of compact object mergers from dynamical interactions with SMBH binaries*, 2019, [MNRAS](#), **490**, 2627
- [6] Liu B., Lai D., **Wang Y.-H.**, *Binary Mergers near a Supermassive Black Hole: Relativistic Effects in Triples*, 2019, [ApJL](#), **883**, L7
- [5] Liu B., Lai D., **Wang Y.-H.**, *Black Hole and Neutron Star Binary Mergers in Triple Systems. II. Merger Eccentricity and Spin-Orbit Misalignment*, 2019, [ApJ](#), **881**, 41
- [4] Perna R., **Wang Y.-H.**, Farr W. M., Leigh N., Cantiello M., *Constraining the Black Hole Initial Mass Function with LIGO/Virgo Observations*, 2019, [ApJL](#), **878**, L1
- [3] **Wang Y.-H.**, Leigh N., Sesana A., Perna R., *Hypervelocity binaries from close encounters with a SMBH-IMBH binary: orbital properties and diagnostics*, 2019, [MNRAS](#), **482**, 3206
- [2] **Wang Y.-H.**, Leigh N., Yuan Y.-F., Perna R., *The fate of close encounters between binary stars and binary supermassive black holes*, 2018, [MNRAS](#), **475**, 4595
- [1] Liu B., **Wang Y.-H.**, Yuan Y.-F., *Modified evolution of stellar binaries from supermassive black hole binaries*, 2017, [MNRAS](#), **466**, 3376

## SELECTED TALKS

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### Invited Talks & Seminars

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|------------------------------------------------------------------------------|-----------|
| • Transient Phenomena and Physical Processes Around Supermassive Black Holes | Oct. 2024 |
| • Graduate Seminar, Nanjing University                                       | Jun. 2023 |
| • AGN Santa Fe Conference, Los Alamos National Lab                           | Mar. 2023 |
| • Graduate Seminar, Stony Brook University                                   | Feb. 2023 |
| • Astro-coffee, IAS, Princeton University                                    | Feb. 2023 |
| • Bahcall Lunch Talk, IAS, Princeton University                              | Feb. 2023 |
| • Graduate Seminar, Georgia Institute of Technology                          | Sep. 2022 |
| • Astrophysics Seminar, University of the Balearic Islands                   | Oct. 2021 |
| • Astronomy Seminar, Universidad de Concepción                               | Sep. 2021 |
| • Astronomy Group Meeting, Cornell University                                | Jun. 2018 |

### Contributed Talks

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|-----------------------------------------------------------------|-----------|
| • 50 years of Binaries and Disks: Lubow@75                      | Apr. 2024 |
| • Anticipating the Rising Tide of Tidal Disruption Events, KITP | Apr. 2024 |
| • 53rd Annual DDA Meeting, CCA, Flatiron Institute              | Apr. 2022 |
| • Planet Formation Group Meeting, CCA, Flatiron Institute       | Oct. 2021 |
| • Astronomy Seminar, Stony Brook University                     | Aug. 2021 |
| • Compact Object Group Meeting, CCA, Flatiron Institute         | Mar. 2021 |
| • Astronomy Seminar, American Museum of Natural History         | Aug. 2018 |

SOFTWARE & RESEARCH PRODUCTS

VegasAfterglow

[GitHub]

Primary Developer | High-speed modeling engine for Gamma-Ray Burst (GRB) afterglows.

- Optimized C++ engine capable of generating light curves in milliseconds, enabling real-time data fitting and massive population studies.
- Supports diverse jet structures and emission mechanisms (Synchrotron & Inverse Compton) to model complex, realistic burst environments.
- Features a Python interface designed for seamless integration with standard fitting tools (like MCMC), allowing researchers to easily analyze observational data.
- *Publication:* Wang et al. (2025), JHEAP [Paper 30].

SpaceHub

[GitHub]

Primary Developer | High-precision few-body gravity integration toolkit.

- Implemented Algorithmic Regularization (AR) and symplectic integrators to handle extreme mass ratios and eccentricities (e.g., SMBHs, chaotic star clusters).
- Designed an easy-to-use API allowing community adoption for diverse dynamical problems.
- *Publication:* Wang et al. (2021), MNRAS [Paper 12].

Secular

[GitHub]

Primary Developer | Fast solver for hierarchical secular dynamics.

- Solves orbit-averaged equations of motion for hierarchical triples/quadruples.
- Highly optimized for studying long-term Lidov-Kozai oscillations and stability limits.

Code Extensions & Modifications

- **Phantom (SPH):** Implemented custom radiative Equation of State (EoS) and modified accretion capture logic to simulate Tidal Disruption Events (TDEs).
- **Athena++ (MHD):** Configured relativistic MHD setups for simulating short GRB jets and their interaction with the ambient medium.

TECHNICAL SKILLS

**Languages:** C/C++, Fortran, Python,  $\text{\LaTeX}$ , HTML, Markdown  
**HPC:** MPI, OpenMP, Slurm workload manager  
**Astrophysics Codes:** Phantom, Athena++, Rebound, SpaceHub

MENTORING

|                                               |                |
|-----------------------------------------------|----------------|
| Graduate Students                             |                |
| Connery Chen, University of Nevada, Las Vegas | 2023 - Present |
| Chaitanya Prasad, Stony Brook University      | 2022 - 2023    |
| Michael Ray, Stony Brook University           | 2021 - 2022    |
| Undergraduate Students                        |                |
| Antonio Frigo, Stony Brook University         | 2021           |
| Robert Serrano, Stony Brook University        | 2019 - 2020    |

PROFESSIONAL SERVICE & OUTREACH

Collaboration Memberships

- Member, *Einstein Probe (EP)* Collaboration2024 - Present
- Member, *SVOM* (Space-based multi-band astronomical Variable Objects Monitor)2024 - Present

**Referee Service:** *ApJ*, *ApJL*, *MNRAS*, *A&A*, *JHEAp*.

**Leadership:**

- Coordinator, NCfA Multi-messenger Meeting 2022 - 2025
- Mentor, Stony Brook Graduate Mentoring Program 2017 - 2019

**Outreach:**

- UNLV Community Engagement Expo 2023
- Astronomy on Tap (Las Vegas): "100 Years of Variable Stars" 2023

## TEACHING EXPERIENCE

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|                                                                        |             |
|------------------------------------------------------------------------|-------------|
| Teaching Assistant, AST-203: <i>Astronomy</i>                          | Spring 2019 |
| Teaching Assistant, AST-248: <i>Search for Life in the Universe</i>    | Fall 2018   |
| Teaching Assistant, PHY-134: <i>Classical Physics Lab II</i>           | Spring 2018 |
| Teaching Assistant, AST-205: <i>Introduction to Planetary Sciences</i> | Fall 2017   |
| Teaching Assistant, <i>Electromagnetism</i>                            | Fall 2014   |

## SELECTED PRESS COVERAGE

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### Evidence of a Past Merger of the Galactic Center Black Hole

- *CBC*: "A cosmic collision 9 billion years ago..." Sept. 2024
- *Phys.org*: "Massive merger: Study reveals evidence..." Sept. 2024

### Floating binary planets from ejections during close stellar encounters

- *Quanta Magazine*: "Rogue Worlds Throw Planetary Ideas Out of Orbit" Nov. 2023
- *Physics World*: "Pairs of rogue planets found wandering..." Oct. 2023

### A planet could have been stolen from the solar system as it formed

- *New Scientist* Feb. 2020

## REFERENCES

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**Rosalba Perna**

Stony Brook University

Email: [rosalba.perna@stonybrook.edu](mailto:rosalba.perna@stonybrook.edu)

**Douglas N. C. Lin**

University of California, Santa Cruz

Email: [lin@ucolick.org](mailto:lin@ucolick.org)

**Nicholas Stone**

University of Wisconsin, Madison

Email: [nstone2@wisc.edu](mailto:nstone2@wisc.edu)

**Ye-Fei Yuan**

University of Science and Technology of China

Email: [yfyuan@ustc.edu.cn](mailto:yfyuan@ustc.edu.cn)

**Bing Zhang**

University of Nevada, Las Vegas

Email: [bing.zhang@unlv.edu](mailto:bing.zhang@unlv.edu)

**Zhaohuan Zhu**

University of Nevada, Las Vegas

Email: [zhaohuan.zhu@unlv.edu](mailto:zhaohuan.zhu@unlv.edu)

**Dong Lai**

Cornell University

Email: [dong@astro.cornell.edu](mailto:dong@astro.cornell.edu)